

BHAWNA PAL

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AI Full Stack Developer | Software Engineer

[GitHub](#) | [LinkedIn](#) | [HackerRank](#) | [Portfolio Website](#)

PROFESSIONAL SUMMARY

I am a software developer with hands-on experience in **Java Full Stack Development, AI/ML, and backend engineering**. Proficient in **Core Java, Advanced Java, Spring Boot, Spring JDBC, React.js, REST APIs, and JWT authentication**. Experienced in building **scalable backend systems, multi-agent AI solutions, and data-driven analytics applications**. Strong understanding of **Agile/Scrum, API integration, and database-driven development**. Focused on delivering **clean, scalable, and reliable solutions to real-world business problems**.

SKILLS

- **Languages:** Java (Proficient), Python (Intermediate), JavaScript (Basic)
- **Frontend:** React.js, HTML, CSS, Streamlit, Tailwind.
- **Backend & APIs:** Spring Boot, Spring JDBC, RESTful APIs, JWT Authentication, LangChain, LangGraph, LLMs.
- **Database & Data Visualization:** SQL, MySQL, NumPy, Pandas, Scikit-learn, MATLAB, Seaborn.
- **Tools & Platforms:** Git, GitHub, Docker, Gradle, IntelliJ IDEA, VS Code, Jupyter, Postman, AWS Services.
- **Practices:** Agile, Scrum.

SOFT SKILLS

Communication | Teamwork & Collaboration | Problem Solving | Time Management | Leadership | Adaptability | Critical Thinking

EXPERIENCE

Junior Engineer Intern – Ironbook AI

June 2025 – September 2025 | Remote

- developed the **INitro - Smart Agentic Decisioning System**, a multi-agent enabling autonomous decision-making and task execution.
- Implemented LLM-based agent workflows using **LangChain and LangGraph**, improving system efficiency and real-time response.
- Integrated APIs and optimized prompt engineering & data pipelines, enhancing accuracy, scalability, and overall system performance.

Machine Learning Intern – Smart Swift Innovation Pvt. Ltd.

August 2024 – September 2024 | Remote

- Developed and deployed machine learning models using supervised and unsupervised techniques on real-world datasets
- Performed **data preprocessing, feature engineering, and visualization** using Python libraries (**NumPy, Pandas, Matplotlib, Scikit-learn**)
- Improved model performance through hyperparameter tuning, evaluation metrics, and iterative optimization techniques.

PROJECTS

ResQNet – Unified Emergency Response & Community Assistance Platform

Java, Spring Boot, Spring JDBC, React.js, Tailwind, JWT, MySQL, Gradle, Docker

- Architected a full-stack emergency response platform using **Java, Spring Boot, Spring Data JPA, and React.js (Vite)**, comprising **40+ backend components** with a scalable relational database architecture.
- Implemented **JWT-based authentication** and **Role-Based Access Control (RBAC)** across **4 user roles** — Users, Volunteers, NGOs, and Admins — ensuring secure and role-specific platform access.
- Engineered a **blood donor compatibility and matching engine** with rule-based matching logic, achieving **95%+ matching accuracy** across supported blood groups.
- Developed an **emergency request state machine** to manage request lifecycles and streamline the routing of urgent assistance requests.
- Built volunteer and NGO **registration and verification workflows**, digitizing onboarding and reducing manual processing effort by approximately **40%**.
- Integrated **RESTful APIs** between the React.js frontend and Spring Boot backend, enabling seamless communication across **15+ API endpoints**.

INitro Smart Agentic Decisioning System

LangChain, LangGraph, LLMs, FastAPI | GitHub link: <https://github.com/BhawnaPal2407/INitro-Smart-Agentic-Decisioning-System>

- Engineered an AI-powered **multi-agent decisioning system** using **LangChain and LangGraph** for customer segmentation, automated campaign generation, and dynamic task execution.
- Designed a **memory-based agent architecture** enabling contextual decision-making and improving response consistency across multi-step.
- Implemented modular AI agents for **customer analysis, campaign generation, and task orchestration**, reducing manual decision-making effort by approximately **60%**.
- Integrated **LLM-powered workflows** with structured prompts and agent coordination, achieving **90%+ task execution accuracy**.

AI-Powered Job & Skill Gap Analyzer

Python 3.9+, Hugging Face, FastAPI, Streamlit | Live: skillfit-chi.vercel.app

- Developed an AI-powered **job matching system** analyzing **50+ skills** against job-role requirements, generating **90%+ accurate match scores** with detailed hit/miss analysis.
- Integrated **Hugging Face models** with **FastAPI**, automating skill-gap analysis and processing **100+ job-skill comparisons** through APIs.
- Built an interactive **Streamlit dashboard** delivering personalized **upskilling recommendations**, reducing manual skill-gap analysis effort by **70%**.

CERTIFICATIONS

***Full Stack Java Development — Tata STRIVE | 100% Merit Scholarship** — Selected through competitive **merit assessment & interview**; trained in **Java, Spring Boot, Spring JDBC, REST APIs, SQL, and full-stack application development**.

***Machine Learning with Python — YHills** | Hands-on training in **ML algorithms, data preprocessing, feature engineering, model evaluation, and data science workflows** using **Python & Scikit-learn**.

EDUCATION

- Bachelor of Technology in Computer Science — Management and Education Research Institute, Sampla, Haryana | October 2023 – June 2026
- Diploma in Electrical Engineering — G.B. Pant Government Engineering College, Okhla, New Delhi | June 2020 – August 2023

EXTRACURRICULAR ACTIVITIES

Freelancer – Event Management: Planned and executed multiple client events end-to-end, strengthening leadership, teamwork, and time management in high-pressure environments.